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INTERNATIONAL DEVELOPMENT ASSOCIATION

INTERNATIONAL MONETARY FUND

ISLAMIC REPUBLIC OF AFGHANISTAN

Joint World Bank-IMF Debt Sustainability Analysis

June 2021

Prepared Jointly by the staffs of the International Development Association (IDA)

and the International Monetary Fund (IMF)

Approved by Marcello Estevão (IDA), Zeine Zeidane and Gavin Gray (IMF)

Islamic Republic of Afghanistan Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Sustainable*
Application of judgement	Yes: The projection period informing mechanical risk signals is extended to 20 years.

This debt sustainability analysis (DSA) update confirms Afghanistan's risk of external and overall debt distress as high—unchanged from the previous DSA conducted in the context of the Extended Credit Facility (ECF) arrangement's approval in November 2020.[1] The macroeconomic framework underlying the update applies minor adjustments to the long-term growth, inflation, fiscal, and external paths of the November 2020 DSA. Mechanical signals in the first decade of projections suggest a moderate risk of debt distress, but the extended 20-year period is used given the projected shift in the financing mix from grants towards debt. During this extended period, the ratio of present value of debt-to-exports repeatedly breaches its threshold under the baseline, justifying the high-risk rating. Maintaining Afghanistan's debt sustainability in the face of substantial downside risks from elevated political uncertainty, faster-than-expected-drop in aid, contingent liabilities, and weather shocks will require sound fiscal policy with prudent deficit levels, revenue mobilization and improved public spending effectiveness, and enhanced debt management. The authorities should also strengthen fiscal risk management capacity to mitigate risks from state-owned corporations (SOCs) and public private partnerships (PPPs).

[1] This DSA was jointly prepared by IMF and World Bank staff.

BASELINE SCENARIO

1. Macroeconomic assumptions underlying this DSA update are broadly in line with those in the previous DSA. GDP growth and inflation are expected to average 4.0 percent and 4.3 percent over the long term. Projections for grants over the long run are slightly lower, by an average of 2 percentage points of GDP, than in the previous DSA, with a commensurately lower primary expenditure, leaving fiscal deficit projections broadly unchanged. Exports and noninterest current account deficits are also projected close to the November 2020 DSA. Afghanistan received SDR 4.8 million debt service relief from the IMF's Catastrophe Containment and Relief Trust (CCRT) in 2020 and is scheduled to receive SDR 2.4 million CCRT relief for debt service due from April 14, 2021 to October 15, 2021. It benefitted from the Debt Service Suspension Initiative in 2020 (0.02 percent of GDP), the authorities requested an extension through June 2021 and are planning to request a further extension for the second half of 2021.

Text Table 1: Macroeconomic Assumptions Comparison Table

	DSA Nov 2020		Current DSA	
	2020-25	2026-40	2021-26	2027-41
Real GDP (%)	2.7	4.0	4.0	4.0
Inflation (GDP, deflator, %)	4.3	4.3	4.1	4.3
Revenue and grants (% of GDP)	27.9	24.6	25.8	23.3
Grants (% GDP)	13.3	6.4	11.2	4.7
Primary expenditure (% GDP)	29.3	26.4	26.9	25.2
Primary deficit (% GDP)	1.3	1.9	1.0	1.9
Exports of G&S (% change)	11.4	6.6	8.6	6.2
Noninterest current account deficit (% GDP)	-7.9	9.1	-7.8	9.0

Sources: Afghan authorities and IMF staff estimates and projections

2. Afghanistan continues to be assessed as having weak debt-carrying capacity. Based on the October 2020 WEO macroeconomic indicators and World Bank's 2019 CPIA (Country Policy and Institutional Assessment), Afghanistan's composite indicator (CI) score is 2.54, below the lower cut-off value of 2.69, indicating a weak debt-carrying capacity rating. The composite score has declined since October 2020 mostly due to the change in global growth assumptions.

3. Public debt used in this DSA does not cover liabilities from SOCs and PPPs. The authorities are developing capacity to systematically collect and record data on financial liabilities of SOCs and under PPPs, including related contingent liabilities. Until such data becomes available, DSAs will continue to use default values for contingent liability shocks. Public debt coverage remains the same as in the November 2020 DSA.

RISK RATING AND VULNERABILITIES

4. Afghanistan's risk of external debt distress remains high. This DSA continues to use a 20-year projection period for the mechanical risk rating considering the significant, if gradual, shift in the financing mix over the long term. A replacement of grants by debt leads to sustained debt accumulation and to large, persistent breaches of the relevant threshold by the PV of external debt-to-exports ratio under the baseline starting 2033 and through the rest of the projection period.[2] Among the liquidity indicators, debt service-to-exports ratio also breaches its threshold in the early 2040s, but the debt service-to-revenue ratio remains low (Figure 1).

5. External debt sustainability is most vulnerable to the export shock, which is the most extreme shock for the PV of debt-to-exports and debt service-to-exports ratios (Table 3). The combination shock—a combination of the growth, primary balance, exports, other flows and depreciation shocks in half their magnitudes—is the most extreme shock for the PV of debt-to-GDP and debt service-to-revenue ratios. These results reflect Afghanistan's high dependence on grants and underscore the need to mobilize domestic revenue and boost exports.

6. Public debt levels are projected to remain low. The PV of total public debt-to-GDP ratio is projected to stay well below the benchmark under the baseline. However, this indicator would breach the threshold under the stress scenario with a growth shock starting 2031. The PV of total public debt-to-revenue ratio and the debt service-to-revenue ratio are most sensitive to the growth shock as well (Figure 2).

7. Afghanistan's overall risk of public debt distress is high. This assessment is based on the application of mechanical signals over the extended period which is necessary given the projected decline in grants and change in the financing mix. Based on these signals over the 20-year horizon, staff assesses the overall risk of debt distress as high.

CONCLUSIONS

8. Sound fiscal policy with prudent deficit levels, revenue mobilization, and improved debt and fiscal risk management are key to safeguarding Afghanistan's debt sustainability. As the economy recovers from the pandemic, the authorities should reverse the 2020-21 fiscal loosening due to the pandemic and step up domestic revenue mobilization. They should rely on highly concessional financing consistent with the ECF-supported program and further strengthen public investment management to finance priority projects with high growth and social returns. Contingent liability risks should be monitored carefully in light of the banking sector vulnerabilities and recent changes to the institutional framework for PPPs. The authorities could continue developing debt management and fiscal risk management capacities with technical assistance from the IMF, World Bank and other partners.

[2] See ¶87 of "Guidance Note on the Bank-Fund Debt Sustainability Framework for Low Income Countries."

AUTHORITIES' VIEWS

9. **The authorities agreed with the conclusions of the DSA update.** They remain committed to fiscal prudence and plan to rely on donor grants and concessional borrowing to ensure debt sustainability, while working to increase revenue and support growth-enhancing investments. Authorities are interested in TA on debt management, including for Sukuk issuance, and are planning to collect information on financial performance on SOCs and build data on public projects, including those executed through PPPs.

Text Table 2: Coverage of Public Debt and Design of Contingent Liability Test

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

1	The country's coverage of public debt	The central government, central bank			
		Default		Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0	percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2	percent of GDP	2.0	
4	PPP	35	percent of PPP stock	0.0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5	percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)			7.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

Text Table 3 : Calculation of the CI Index

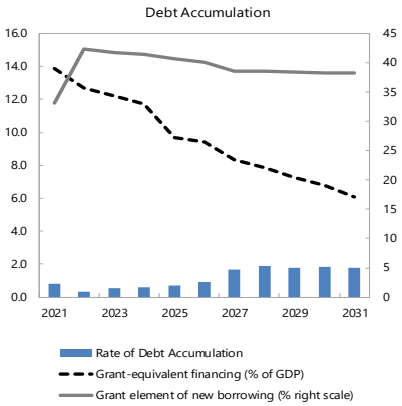
Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	2.6933	1.04	41%
Real growth rate (in percent)	2.719	2.2932	0.06	2%
Import coverage of reserves (in percent)	4.052	57.9628	2.35	92%
Import coverage of reserves^2 (in percent)	-3.990	33.5968	-1.34	-53%
Remittances (in percent)	2.022	1.9716	0.04	2%
World economic growth (in percent)	13.520	2.9280	0.40	16%
CI Score			2.54	100%
CI rating			Weak	

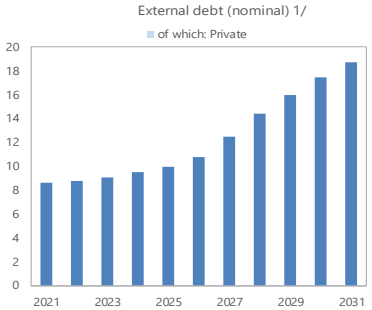
Table 1. Islamic Republic of Afghanistan: External Debt Sustainability Framework, Baseline Scenario, 2020–41

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
External debt (nominal) 1/	**7.5**	**8.6**	**8.7**	**9.1**	**9.5**	**10.0**	**10.8**	**18.7**	**17.5**	**6.8**	**12.3**
of which: public and publicly guaranteed (PPG)	**7.5**	**8.6**	**8.7**	**9.1**	**9.5**	**10.0**	**10.8**	**18.7**	**17.5**	**6.8**	**12.3**
Change in external debt	1.4	1.1	0.1	0.3	0.4	0.5	0.8	1.3	-1.5		
Identified net debt-creating flows	**-11.5**	**-11.1**	**-9.8**	**-9.3**	**-8.9**	**-7.2**	**-5.5**	**5.6**	**6.7**	**-10.7**	**-3.9**
Non-interest current account deficit	**-11.4**	**-10.7**	**-8.9**	**-8.4**	**-8.0**	**-6.1**	**-4.4**	**8.1**	**11.0**	**-10.1**	**-2.4**
Deficit in balance of goods and services	27.8	28.8	28.0	27.0	26.2	25.0	24.0	23.1	21.7	**33.4**	**25.1**
Exports	7.5	7.9	8.1	8.4	9.1	9.4	9.9	9.7	9.6		
Imports	35.3	36.7	36.1	35.4	35.2	34.4	33.9	32.8	31.3		
Net current transfers (negative = inflow)	-38.2	-38.6	-35.9	-34.2	-32.9	-29.9	-27.2	-14.3	-10.4	**-42.8**	**-26.6**
of which: official	-35.7	-35.8	-32.9	-31.0	-29.5	-26.3	-23.3	-9.5	-3.5		
Other current account flows (negative = net inflow)	-1.0	-0.9	-1.0	-1.2	-1.2	-1.2	-1.2	-0.7	-0.4	**-0.8**	**-1.0**
Net FDI (negative = inflow)	**0.1**	**-0.2**	**-0.5**	**-0.6**	**-0.6**	**-0.7**	**-0.7**	**-2.0**	**-3.8**	**-0.4**	**-1.0**
Endogenous debt dynamics 2/	**-0.2**	**-0.2**	**-0.3**	**-0.3**	**-0.3**	**-0.4**	**-0.4**	**-0.5**	**-0.5**		
Contribution from nominal interest rate	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2		
Contribution from real GDP growth	0.1	-0.2	-0.4	-0.4	-0.4	-0.4	-0.4	-0.7	-0.7		
Contribution from price and exchange rate changes	-0.4	...	...	...	...	...	...	...	...		
Residual 3/	**12.9**	**12.2**	**9.9**	**9.6**	**9.4**	**7.6**	**6.3**	**-4.3**	**-8.2**	**10.6**	**4.9**
of which: exceptional financing	-1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
CCRT Debt Relief	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**3.5**	**4.3**	**4.4**	**4.7**	**5.2**	**5.6**	**6.2**	**11.9**	**13.0**		
PV of PPG external debt-to-exports ratio	**47.4**	**54.3**	**54.6**	**56.0**	**56.9**	**59.3**	**63.2**	**122.8**	**135.5**		
PPG debt service-to-exports ratio	**2.4**	**2.7**	**2.5**	**2.7**	**2.9**	**2.9**	**2.9**	**4.9**	**10.6**		
PPG debt service-to-revenue ratio	**1.6**	**1.8**	**1.5**	**1.6**	**1.7**	**1.7**	**1.8**	**2.6**	**5.1**		
Gross external financing need (Million of U.S. dollars)	-2194.2	-2148.6	-1942.2	-1923.8	-1880.0	-1535.4	-1192.6	2223.5	5097.4		
Key macroeconomic assumptions											
Real GDP growth (in percent)	-1.9	2.7	4.5	4.5	4.0	4.0	4.0	4.0	4.0	**3.8**	**4.0**
GDP deflator in US dollar terms (change in percent)	6.9	-1.0	0.1	-0.3	-1.4	0.9	-0.2	3.2	2.1	**-0.9**	**0.9**
Effective interest rate (percent) 4/	0.7	0.5	0.5	0.5	0.5	-0.2	0.0	0.8	1.3	**0.4**	**0.4**
Growth of exports of G&S (US dollar terms, in percent)	-2.2	7.5	7.2	8.8	10.3	8.6	8.9	6.3	5.5	**-1.8**	**7.5**
Growth of imports of G&S (US dollar terms, in percent)	-5.3	5.8	2.8	2.4	2.0	2.3	2.3	5.9	5.5	**2.8**	**4.3**
Grant element of new public sector borrowing (in percent)	...	33.2	42.3	41.7	41.5	40.6	40.1	38.3	42.1	**...**	**39.2**
Government revenues (excluding grants, in percent of GDP)	11.4	12.3	13.4	14.2	15.7	16.0	16.1	18.0	19.7	**11.4**	**16.0**
Aid flows (in Million of US dollars) 5/	3271.2	2734.6	2703.3	2713.9	2704.6	2387.5	2448.2	2447.1	2288.8		
Grant-equivalent financing (in percent of GDP) 6/	...	13.9	12.7	12.2	11.7	9.7	9.4	6.1	3.7	**...**	**9.6**
Grant-equivalent financing (in percent of external financing) 6/	...	94.2	97.6	96.0	95.2	93.5	91.5	78.9	94.2	**...**	**88.3**
Nominal GDP (Million of US dollars)	19,786	20,116	21,056	21,952	22,505	23,622	24,516	33,666	62,197		
Nominal dollar GDP growth	4.8	1.7	4.7	4.3	2.5	5.0	3.8	7.3	6.2	**2.9**	**5.0**
Memorandum items:											
PV of external debt 7/	3.5	4.3	4.4	4.7	5.2	5.6	6.2	11.9	13.0		
In percent of exports	47.4	54.3	54.6	56.0	56.9	59.3	63.2	122.8	135.5		
Total external debt service-to-exports ratio	2.4	2.7	2.5	2.7	2.9	2.9	2.9	4.9	10.6		
PV of PPG external debt (in Million of US dollars)	699.5	861.9	928.0	1036.4	1161.6	1314.5	1525.5	4014.5	8080.5		
(PVt-PVt-1)/GDPt-1 (in percent)		0.8	0.3	0.5	0.6	0.7	0.9	1.8	-0.1		
Non-interest current account deficit that stabilizes debt ratio	-12.8	-11.8	-9.0	-8.8	-8.4	-6.6	-5.2	6.9	12.4		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

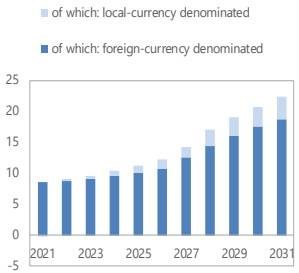
Table 2. Islamic Republic of Afghanistan: Public Sector Debt Sustainability Framework, Baseline Scenario, 2020–41

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**7.5**	**8.6**	**9.0**	**9.6**	**10.5**	**11.2**	**12.2**	**22.3**	**22.5**	**7.6**	**14.0**
of which: external debt	**7.5**	**8.6**	**8.7**	**9.1**	**9.5**	**10.0**	**10.8**	**18.7**	**17.5**	**6.8**	**12.3**
Change in public sector debt	1.4	1.1	0.4	0.6	0.8	0.8	1.0	1.7	-1.7		
Identified debt-creating flows	**2.2**	**2.3**	**0.6**	**0.3**	**0.3**	**0.3**	**0.4**	**1.0**	**0.4**	**0.3**	**0.8**
Primary deficit	**2.2**	**2.4**	**1.1**	**0.6**	**0.6**	**0.7**	**0.8**	**2.0**	**1.3**	**0.6**	**1.4**
Revenue and grants	26.1	25.7	25.9	26.1	27.0	25.2	25.0	23.1	23.2	**25.8**	**25.0**
of which: grants	14.8	13.4	12.5	11.9	11.3	9.2	8.8	5.1	3.5		
Primary (noninterest) expenditure	28.4	28.1	27.1	26.7	27.6	25.9	25.7	25.1	24.5	**26.4**	**26.4**
Automatic debt dynamics	**-0.1**	**0.0**	**-0.2**	**-0.4**	**-0.3**	**-0.4**	**-0.4**	**-1.0**	**-0.9**		
Contribution from interest rate/growth differential	0.0	-0.3	-0.5	-0.5	-0.5	-0.6	-0.6	-0.9	-0.9		
of which: contribution from average real interest rate	-0.1	-0.1	-0.1	-0.1	-0.1	-0.2	-0.2	-0.1	0.0		
of which: contribution from real GDP growth	0.1	-0.2	-0.4	-0.4	-0.4	-0.4	-0.4	-0.8	-0.9		
Contribution from real exchange rate depreciation	-0.1	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**-0.1**	**-0.3**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.1**	**0.0**
Privatization receipts (negative)	0.0	-0.1	-0.3	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-0.8**	**-0.9**	**0.0**	**0.4**	**0.7**	**0.6**	**0.8**	**0.6**	**-2.1**	**-0.3**	**0.6**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**3.6**	**4.4**	**4.8**	**5.4**	**6.2**	**6.9**	**7.8**	**15.7**	**18.2**		
PV of public debt-to-revenue and grants ratio	**13.6**	**17.0**	**18.7**	**20.8**	**22.9**	**27.5**	**31.1**	**68.1**	**78.3**		
Debt service-to-revenue and grants ratio 3/	**0.7**	**0.9**	**0.8**	**0.9**	**1.1**	**2.2**	**2.4**	**8.0**	**12.3**		
Gross financing need 4/	2.4	2.5	1.1	0.9	0.9	1.3	1.4	3.9	4.2		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	-1.9	2.7	4.5	4.5	4.0	4.0	4.0	4.0	4.0	**3.8**	**4.0**
Average nominal interest rate on external debt (in percent)	0.7	0.5	0.5	0.5	0.5	-0.2	0.0	0.8	1.3	**0.4**	**0.4**
Average real interest rate on domestic debt (in percent)	-2.3	-4.2	2.4	2.9	2.9	2.9	2.9	2.5	2.9	**-3.4**	**2.0**
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.0	...	...	...	...	...	...	...	...	**3.4**	**...**
Inflation rate (GDP deflator, in percent)	2.4	4.4	4.5	4.0	4.0	4.0	4.0	4.4	4.0	**4.1**	**4.3**
Growth of real primary spending (deflated by GDP deflator, in percent)	2.6	1.7	0.6	3.1	7.5	-2.3	3.4	3.9	3.6	**7.5**	**2.9**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	0.8	1.3	0.7	0.1	-0.2	0.0	-0.2	0.3	3.0	**0.7**	**0.1**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/



Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

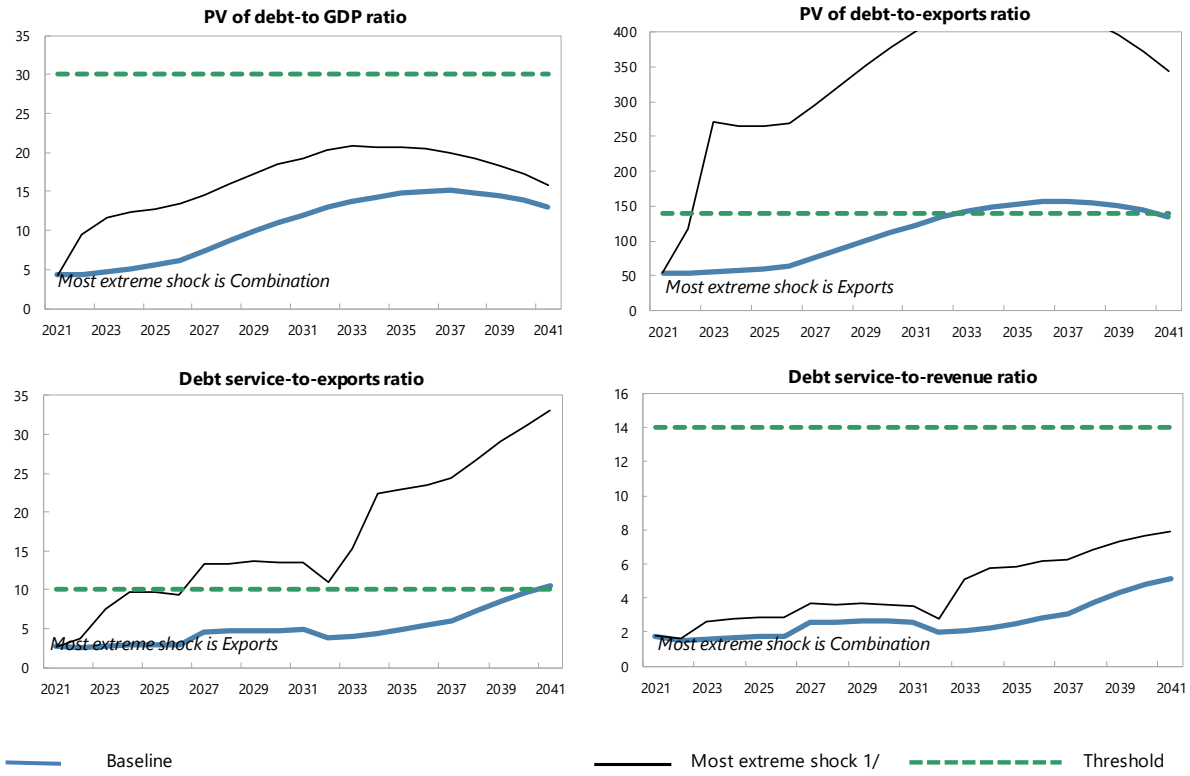
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Islamic Republic of Afghanistan: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2021–41



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.2%	1.5%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	19	20
Avg. grace period	9	10

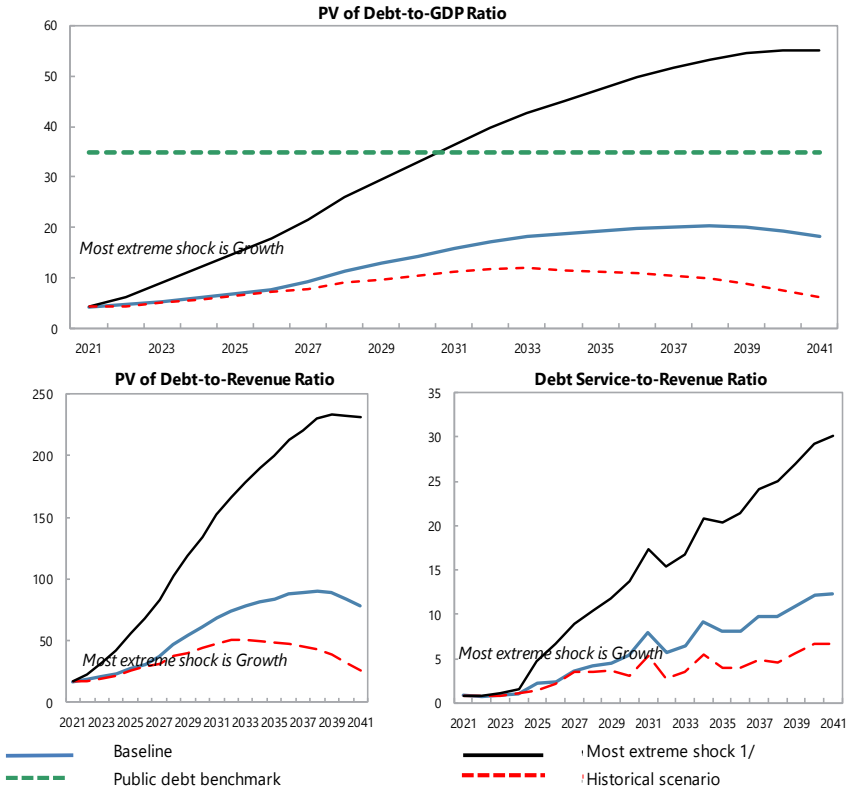
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Islamic Republic of Afghanistan: Indicators of Public Debt Under Alternative Scenarios, 2021–41



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	59%	59%
Domestic medium and long-term	41%	41%
Domestic short-term	0%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.2%	1.5%
Avg. maturity (incl. grace period)	19	20
Avg. grace period	9	10
Domestic MLT debt		
Avg. real interest rate on new borrowing	2.7%	2.7%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	0%	0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Islamic Republic of Afghanistan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021–31

(In percent)

	Projections										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio											
Baseline	4.3	4.4	4.7	5.2	5.6	6.2	7.4	8.7	9.9	11.0	11.9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	4.3	3.8	3.2	2.4	0.5	-2.2	-6.4	-12.0	-19.0	-27.4	-37.1
B. Bound Tests											
B1. Real GDP growth	4.3	4.8	5.5	6.1	6.5	7.3	8.6	10.2	11.6	12.9	14.0
B2. Primary balance	4.3	4.8	5.6	6.0	6.6	7.4	8.5	9.9	11.1	12.2	13.1
B3. Exports	4.3	6.3	9.9	10.4	10.8	11.5	12.5	13.8	15.0	16.0	16.8
B4. Other flows 2/	4.3	7.6	11.1	11.7	12.0	12.7	13.7	14.9	16.0	17.0	17.7
B6. One-time 30 percent nominal depreciation	4.3	5.6	-0.1	0.3	0.9	1.7	3.3	5.1	6.7	8.2	9.5
B6. Combination of B1-B5	4.3	9.5	11.7	12.3	12.7	13.5	14.6	16.0	17.3	18.4	19.3
C. Tailored Tests											
C1. Combined contingent liabilities	4.3	7.0	7.4	7.9	9.2	9.9	11.0	12.5	13.7	14.8	15.7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**
PV of debt-to-exports ratio											
Baseline	54.3	54.6	56.0	56.9	59.3	63.2	75.0	88.5	100.7	112.4	122.8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	54.3	47.0	37.9	26.5	5.4	-22.7	-65.4	-122.6	-193.1	-280.1	-382.4
B. Bound Tests											
B1. Real GDP growth	54.3	54.6	56.0	56.9	59.3	63.2	75.0	88.5	100.7	112.4	122.8
B2. Primary balance	54.3	59.8	66.0	66.6	70.1	75.0	86.8	100.6	113.1	124.6	135.0
B3. Exports	54.3	117.1	**270.3**	**264.8**	**265.2**	**269.0**	**294.7**	**324.5**	**351.6**	**377.5**	**400.2**
B4. Other flows 2/	54.3	94.1	132.0	128.7	128.2	129.0	139.4	**151.7**	**162.8**	**173.4**	**182.7**
B6. One-time 30 percent nominal depreciation	54.3	54.6	-1.4	2.7	7.4	13.5	26.5	40.9	53.8	66.3	77.7
B6. Combination of B1-B5	54.3	138.4	122.7	**197.7**	**197.4**	**199.4**	**216.9**	**237.3**	**255.9**	**273.6**	**289.1**
C. Tailored Tests											
C1. Combined contingent liabilities	54.3	87.1	87.7	87.4	97.9	100.5	111.9	127.7	139.4	**150.7**	**161.4**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**
Debt service-to-exports ratio											
Baseline	2.7	2.5	2.7	2.9	2.9	2.9	4.6	4.7	4.8	4.8	4.9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	2.7	2.5	2.6	2.5	2.4	1.8	3.0	2.0	0.7	-1.1	-3.0
B. Bound Tests											
B1. Real GDP growth	2.7	2.5	2.7	2.9	2.9	2.9	4.6	4.7	4.8	4.8	4.9
B2. Primary balance	2.7	2.5	2.8	3.1	3.1	3.1	4.8	4.9	5.0	5.0	5.1
B3. Exports	2.7	3.8	7.5	9.8	9.7	9.4	13.3	13.4	13.7	13.6	13.6
B4. Other flows 2/	2.7	2.5	3.5	4.5	4.4	4.3	5.9	5.9	6.0	6.0	6.0
B6. One-time 30 percent nominal depreciation	2.7	2.5	2.7	1.7	1.8	1.8	3.6	3.7	3.9	3.9	4.0
B6. Combination of B1-B5	2.7	3.2	6.4	7.1	7.0	6.8	9.5	9.5	9.7	9.6	9.6
C. Tailored Tests											
C1. Combined contingent liabilities	2.7	2.5	3.4	3.6	3.6	3.7	5.4	5.4	5.6	5.6	5.6
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**
Debt service-to-revenue ratio											
Baseline	1.8	1.5	1.6	1.7	1.7	1.8	2.6	2.6	2.7	2.6	2.6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	1.8	1.5	1.5	1.5	1.4	1.1	1.7	1.1	0.4	-0.6	-1.6
B. Bound Tests											
B1. Real GDP growth	1.8	1.6	1.9	2.0	2.0	2.1	3.0	3.0	3.1	3.1	3.1
B2. Primary balance	1.8	1.5	1.7	1.8	1.8	1.9	2.7	2.7	2.8	2.8	2.7
B3. Exports	1.8	1.5	1.9	2.5	2.5	2.5	3.3	3.2	3.3	3.2	3.2
B4. Other flows 2/	1.8	1.5	2.1	2.6	2.6	2.6	3.3	3.3	3.4	3.3	3.2
B6. One-time 30 percent nominal depreciation	1.8	1.9	2.0	1.2	1.3	1.4	2.5	2.6	2.7	2.7	2.7
B6. Combination of B1-B5	1.8	1.6	2.6	2.8	2.8	2.8	3.7	3.6	3.7	3.6	3.6
C. Tailored Tests											
C1. Combined contingent liabilities	1.8	1.5	2.0	2.1	2.1	2.2	3.0	3.0	3.1	3.0	3.0
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Includes official and private transfers and FDI.

Table 4. Islamic Republic of Afghanistan: Sensitivity Analysis for Key Indicators of Public Debt, 2021–31 (In percent)

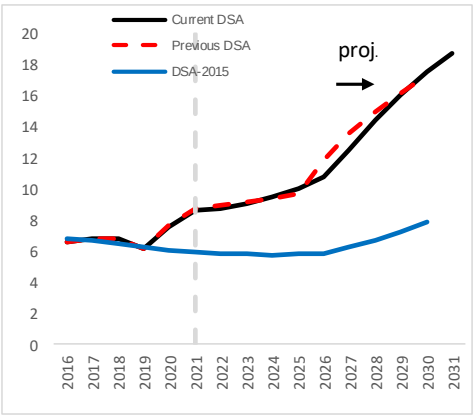
	Projections										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of Debt-to-GDP Ratio											
Baseline	4.4	4.8	5.4	6.2	6.9	7.8	9.3	11.5	13.0	14.3	15.7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	4	4	5	6	6	7	8	9	10	11	11
B. Bound Tests											
B1. Real GDP growth	4	6	9	12	15	18	22	26	30	33	**36**
B2. Primary balance	4	6	7	8	9	9	11	13	14	16	17
B3. Exports	4	7	10	11	12	13	14	16	18	19	20
B4. Other flows 2/	4	8	12	13	14	14	16	18	19	20	22
B6. One-time 30 percent nominal depreciation	4	5	5	5	5	5	5	6	6	7	7
B6. Combination of B1-B5	4	5	6	6	7	8	10	12	14	15	17
C. Tailored Tests											
C1. Combined contingent liabilities	4	10	11	11	12	12	14	16	17	18	20
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**
PV of Debt-to-Revenue Ratio											
Baseline	17.0	18.7	20.8	22.9	27.5	31.1	37.6	47.2	54.3	60.5	68.1
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	17.0	17.1	19.2	21.4	25.6	28.7	31.4	37.2	40.1	44.1	47.6
B. Bound Tests											
B1. Real GDP growth	17.0	22.6	32.6	41.9	55.9	67.5	83.1	102.1	118.8	133.9	151.9
B2. Primary balance	17.0	22.0	27.4	29.2	33.8	37.2	43.5	53.0	60.0	66.1	73.7
B3. Exports	17.0	25.7	40.0	41.5	47.3	51.1	57.3	66.8	73.9	79.9	87.5
B4. Other flows 2/	17.0	31.4	46.1	47.4	53.6	57.5	63.7	73.0	80.2	86.1	93.7
B6. One-time 30 percent nominal depreciation	17.0	21.2	19.4	18.5	19.1	19.7	21.2	25.6	27.7	29.4	32.0
B6. Combination of B1-B5	17.0	20.8	23.0	23.8	28.8	32.9	39.9	49.8	57.3	63.7	71.6
C. Tailored Tests											
C1. Combined contingent liabilities	17.0	39.6	41.3	42.5	46.2	50.0	56.2	64.9	72.1	78.0	85.5
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	0.9	0.8	0.9	1.1	2.2	2.4	3.6	4.2	4.4	5.4	8.0
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2041 1/	0.9	0.8	0.9	1.0	1.5	2.2	3.5	3.5	3.7	3.1	5.4
B. Bound Tests											
B1. Real GDP growth	0.9	0.8	1.2	1.6	4.7	6.7	8.9	10.5	11.8	13.7	17.4
B2. Primary balance	0.9	0.8	1.1	1.4	3.9	4.1	3.9	4.9	5.2	5.6	8.3
B3. Exports	0.9	0.8	1.1	1.5	2.7	2.8	4.0	4.6	4.8	5.7	8.3
B4. Other flows 2/	0.9	0.8	1.2	1.6	2.8	2.9	4.2	4.7	5.0	5.9	8.4
B6. One-time 30 percent nominal depreciation	0.9	0.9	1.1	1.1	2.2	0.8	3.6	4.1	4.2	5.0	7.4
B6. Combination of B1-B5	0.9	0.8	0.9	1.1	2.3	2.4	4.0	4.6	4.9	6.0	8.6
C. Tailored Tests											
C1. Combined contingent liabilities	0.9	0.8	1.8	2.0	12.0	3.3	4.5	8.0	5.2	6.1	9.7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
2/ Includes official and private transfers and FDI.

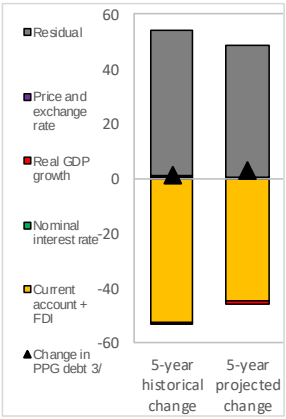
Figure 3. Islamic Republic of Afghanistan: Drivers of Debt Dynamics—Baseline Scenario

External debt

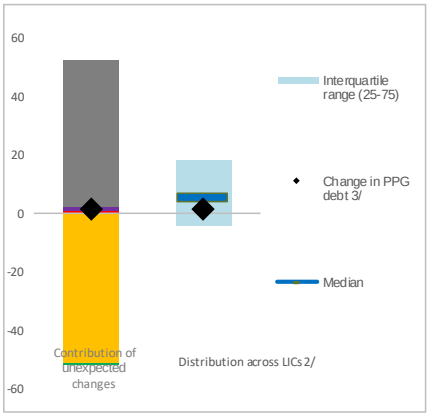
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

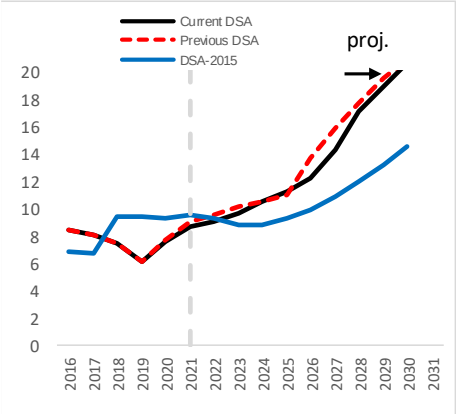


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

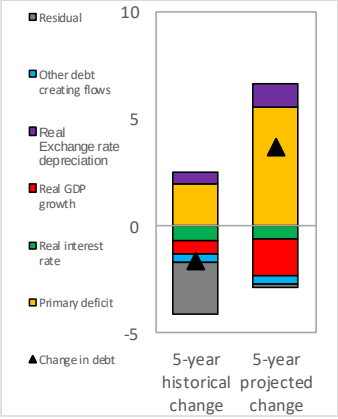


Public debt

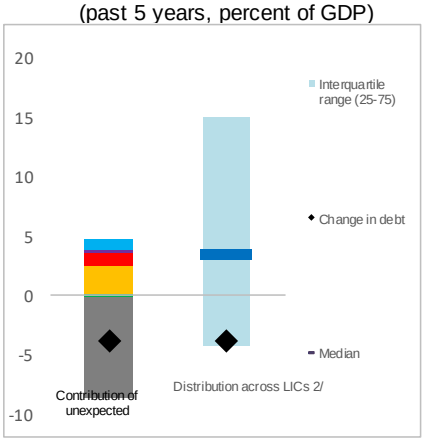
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



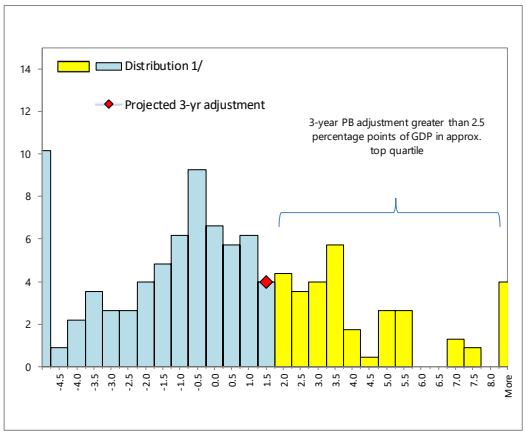
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

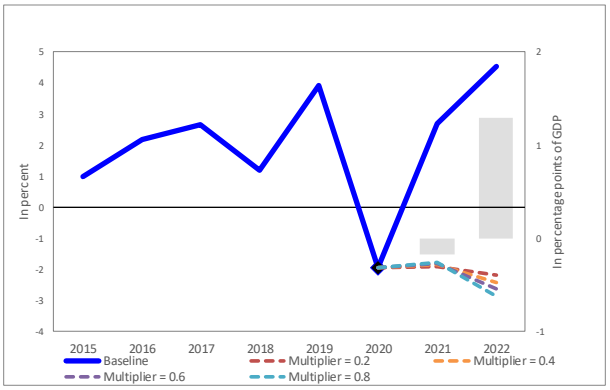
Figure 4. Islamic Republic of Afghanistan: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



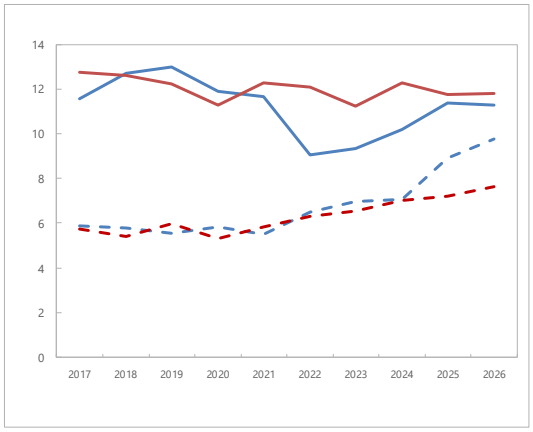
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



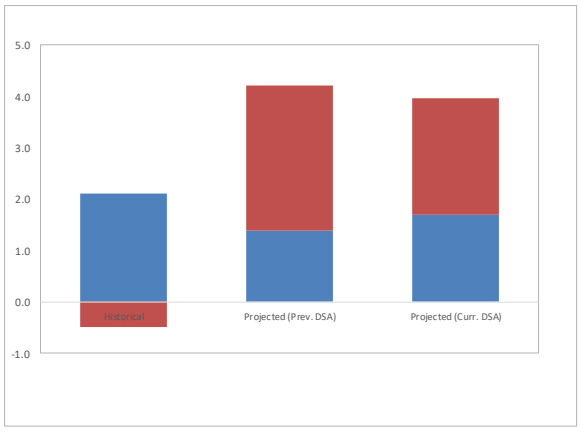
1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates 1/
(% of GDP)



Gov. Invest. - Prev. DSA
Gov. Invest. - Current DSA
Priv. Invest. - Prev. DSA
Priv. Invest. - Current DSA

Contribution to Real GDP growth
(percent, 5-year average)



Contribution of other factors
Contribution of government capital

1/ Historical private and public investment rates are different from previous DSA due to data revisions.